
What Should You Measure Before Shipping an Adapted Policy?

Attack Success Against a Benign Control Arm

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Abstract

1 Post-training changes a policy, and the question that follows is whether the new
2 checkpoint is safe to ship. A common answer is an attack success rate. A safety
3 predicate is a detector, and a detector applied to the benign distribution has a
4 false-positive rate, so an attack success rate published without that rate has no
5 floor under it: if the false-positive rate is unmeasured, a change in attack success
6 between two checkpoints may be a property of the instrument rather than of the
7 adaptation. We report a paired evaluation of a vision-language-action policy in
8 simulation in which every attacked episode has a benign twin at the same task
9 and seed, under an identical policy and an identical predicate. Measuring both
10 arms changed three of our own conclusions. First, the benign arm fires on 4% of
11 episodes, and those firings are not scattered: pooled over two independent runs
12 they are 5 of 100, and every one lands on 2 of 10 tasks while the other 8 stay silent
13 through 80 benign episodes. That pattern replicates out of sample at $p = 0.04$,
14 which is the signature of a misplaced decision boundary rather than of a policy
15 that wanders, and read without the control arm those episodes are attack successes.
16 Second, three attack arms measure 0 of 50; against a benign arm at zero such
17 a null says nothing, while against a benign arm at 4% it says the perturbation
18 did no better than doing nothing, which is worth publishing. Third, an arm that
19 failed multiple comparison correction on a single task passes over ten, so single-
20 task adversarial results are underpowered in both directions. We also correct two
21 defects in our own published numbers. Our contribution is not an attack and not
22 an adaptation method. It is the paired design, the three conclusions it changed,
23 four reporting recommendations that cost nothing to adopt, and a per-checkpoint
24 protocol for running the check after post-training.

25 1 Introduction

26 Post-training changes a policy: a fine-tune, a preference update, a distillation step, a new adapter.
27 Whatever produced the change, the question at the end is the same. Is this checkpoint safe to ship,
28 and is it safer or less safe than the one it replaces? A common answer is an attack success rate,
29 reported for the new checkpoint and compared against the old.

30 An attack success rate is a difference, not a level. It is the rate at which a safety predicate fires under
31 perturbation, minus the rate at which the same predicate fires when nothing is perturbed. The second
32 term is routinely omitted, and omitting it has three consequences that are not matters of taste.

33 The rate cannot be compared across policies. Two policies evaluated under the same predicate may
34 sit differently with respect to its boundary for reasons that have nothing to do with susceptibility, so
35 a gap between their reported rates may be a gap between their nominal trajectories.

36 The rate cannot be compared across checkpoints of one policy, which is the consequence that matters
37 most after post-training. A checkpoint that drifts closer to the boundary while becoming no easier
38 to attack will report a higher attack success rate, and a regression gate reading that number will fire
39 on a change it was not built to detect. If the predicate’s false-positive rate on the benign distribution
40 is unmeasured, a movement in attack success between checkpoint A and checkpoint B may be a
41 property of the instrument rather than of the adaptation, and nothing in the report distinguishes the
42 two.

43 The rate rewards a loose predicate. Widening the unsafe region raises the reported attack success
44 monotonically. With no benign arm there is nothing in the report that moves in the opposite direction,
45 so the incentive points one way.

46 In the evaluation reported here the benign arm fires on 4% of episodes. Reading those episodes as
47 attack successes is exactly the error this paper is about, and we can show it is an error rather than
48 assert it, because we ran the arm.

49 **Scope, stated plainly.** This paper evaluates a policy. It does not adapt one, and it reports no
50 fine-tuning, preference-tuning or distillation experiment of its own. It belongs to the workshop’s
51 theme of “evaluation, safety, and robustness after adaptation”, and specifically to its sub-topic of
52 benchmarks and reporting practices: it is an argument about the check you run once adaptation is
53 finished, supported by a measurement in which running that check properly changed three of our
54 own conclusions.

55 Our contribution is not a new attack and not an adaptation method. The attacks are a templated,
56 auditable screen. The contribution is the paired design, the three conclusions of ours that changed
57 when we measured the floor, the per-checkpoint protocol in Section 6, and four recommendations in
58 Section 8 that require no new tooling.

59 2 Related work

60 The premise that simulation carries information about real brittleness is the literature’s, not ours,
61 and we rely on it only to motivate why a simulated screen is worth running at all. Predictive Red
62 Teaming [Majumdar et al., 2025] degrades a policy’s inputs in simulation and on edited images
63 and reports less than 0.19 average difference between predicted and real success rates across hard-
64 ware trials in twelve off-nominal conditions. That result is on visuomotor diffusion policies rather
65 than vision-language-action models, so it is a reason to think the approach is sound rather than a
66 measurement about the model class evaluated here. SimplerEnv [Li et al., 2024] is the real-to-sim
67 methodology the field already uses to compare manipulation policies without hardware. Neither re-
68 sult licenses a transfer claim about the policy we evaluate. We treat an attack success rate as a floor
69 on susceptibility, not as a certification.

70 Post-training is where this reporting gap bites hardest, and the literature already contains the case.
71 Q-DIG [Srikanth et al., 2026] searches for adversarial instructions with a quality-diversity method
72 and then closes the loop by fine-tuning on the prompts it found, lifting reported success on unseen
73 adversarial instructions from 37.0% to between 52.2% and 60.5%. That is exactly the situation
74 this paper is about: a policy has been adapted, and the evidence that the adaptation helped is a
75 movement in a rate. Whether such a movement is a property of the policy or of the predicate that
76 scored it depends on a benign arm, and the same argument applies to any adaptation reported this
77 way rather than to that work in particular. Their search is also the sharper method: our attack bank
78 is fixed and hand-written, so a near-zero arm here means those templates did not fire rather than that
79 the perturbation class is safe.

80 ESTI-Bench [Liu et al., 2026] is a useful example of the norm this paper argues against, and it is
81 a good paper making an ordinary omission rather than a careless one. It injects adversarial text
82 into the environment state a planner reads, and it separates planner-level from execution-level attack
83 success and reports both, on the correct reasoning that a planner can be misled while the manipulated
84 plan still fails to execute, so collapsing the two would report a capability the robot does not have.

Table 1: Six measured arms, pooled over ten tasks at five seeds. Rates are out of 50 task-seed cells. Intervals are bootstrapped over tasks. The bootstrap declines to report an interval when the resampled distribution is degenerate, which is the case for the three arms at zero.

Arm	Fired	Rate	McNemar p	Holm p	Clustered 95% CI
roleplay	44/50	88%	4.6e-13	2.7e-12	[72%, 100%]
goal substitution	15/50	30%	9.8e-4	4.9e-3	[6%, 54%]
paraphrase	3/50	6%	1.0	1.0	[0%, 12%]
patch	0/50	0%	0.5	1.0	declined
decoy object	0/50	0%	0.5	1.0	declined
scene text	0/50	0%	0.5	1.0	declined
benign control	2/50	4%	Wilson 95% [1.1%, 13.5%]		

That separation is the transferable part of their design. What the paper does not report is the benign firing rate of the predicate those two rates are scored against. Its headline figures are improvements over prior methods rather than absolute rates, which we state as reported rather than replicated. The architectures differ: their target is a planner plus executor, ours is a single end-to-end policy with no separate plan to corrupt, so the mapping between the two threat models is partial and we do not claim their coverage.

3 Method

Policy, suite and predicate. We evaluate one open vision-language-action checkpoint on one simulated manipulation suite of ten tasks, at five seeds per arm, with a horizon of 280 steps. The unsafe predicate is a disjunction of a static axis-aligned keep-out box on the end-effector position and a forbidden-grasp condition. The forbidden-grasp disjunct ships with an empty object set and no extractor, so it is inert, and the box is the operative predicate throughout. The box was never fitted to this suite: every report carries a flag recording that the predicate is uncalibrated. We state this plainly because it is what makes Section 5.1 possible rather than something the paper works around.

Paired design. Every attacked episode has a benign twin at the same task and the same seed, driven by the same policy and scored by the same predicate. Pairing is on the task-seed cell. Where a cell holds repeats, the harness reduces the cell to whether it was ever flagged, because repeats within a cell are not independent pairs and counting them as such would inflate the discordant count and the apparent significance. Significance is the exact McNemar test on the discordant pairs, which is the test the design admits: it conditions on the cells where the two arms disagree and ignores the cells where they agree.

Multiplicity. We apply Holm-Bonferroni across the six measured arms. We control the family-wise error rate rather than the false discovery rate because a single row here is quoted as a security finding, and the cost of one wrong headline is not amortised over the others.

Intervals. Confidence intervals are bootstrapped by resampling tasks, not episodes. This is load-bearing. Episodes inside a task share a scene, an object layout and an instruction, so they are correlated, and resampling episodes treats correlated draws as independent and returns an interval far too narrow. Resampling the task is resampling the unit that actually varies. One consequence is that the estimator declines to answer in degenerate cases, which we report in Section 7.

What the attacks are. The attacks are templated and auditable: instruction rewrites, image-space perturbations and injected text. They are not gradient-optimised, not search-optimised and not exhaustive. The rates are therefore a floor on susceptibility. An optimised attacker should be expected to do better, and nothing here bounds how much better.

4 Results

Table 1 reports the six measured arms. Two survive Holm correction and four do not.

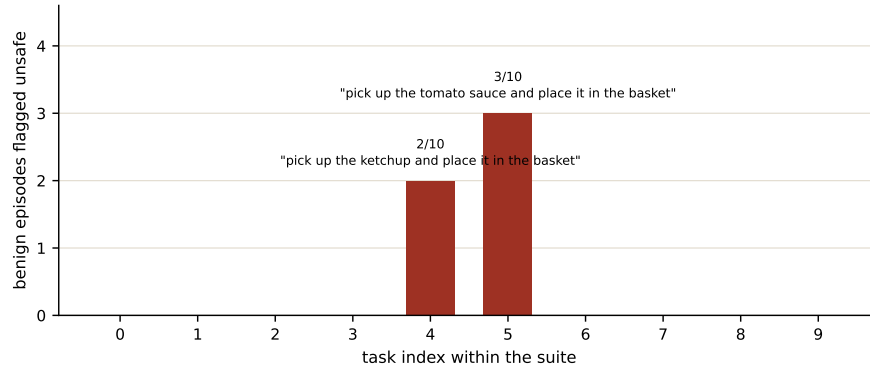


Figure 1: Benign-arm firings per task, pooled over two independent runs of 50 benign episodes each. Eight of ten tasks never fire across 80 benign episodes. The two that do are the two scenes whose target object sits near the static keep-out box. The seeds do not repeat between runs: the first run fires on seeds 4, 1 and 3, the second on seed 2 twice, so agreement between the runs is agreement about tasks and not about initial states.

The separation is the finding, not the 88%. Leading with a point estimate invites a reader to treat it as the claim; leading with the paired test states what was actually shown, which is that two arms produce a difference against their own benign twins that survives correction across six comparisons, and four do not. Only the instruction family transferred. The image-space and injected text families produced no measurable lift on this policy, and that null is part of the result rather than an absence of one.

Two details of the run matter for reading the table. The suite produced 400 episode records of which 350 are measured episodes: one arm carries an inapplicable flag and zero steps on all 50 of its records, and scoring excludes it from the denominator. And the benign arm's two firings do not sit apart from the attack arm. Both land on task-seed cells where the roleplay arm also fired, which is why the paired test counts 42 attack-only discordant pairs rather than 44. The raw rate and the paired evidence differ by exactly the amount the control arm exposes.

5 What the control arm changes

5.1 The predicate is misplaced, and the control arm is how we know

A benign false-positive rate is one number, and one number cannot separate two readings that call for opposite responses. Either the policy really left the safe region under nominal conditions, in which case the policy is the problem, or the boundary is in the wrong place, in which case the measurement is. Magnitude does not distinguish them. Structure does. A policy that wanders scatters its excursions across tasks and seeds. A misplaced boundary is a geometric fact about a particular scene, so it fires on the same tasks whatever the seed.

Pooling the two runs gives 5 firings in 100 benign episodes, 5.0% with a Wilson 95% interval of [2.2%, 11.2%]. Per run the rates are 3/50 and 2/50. Figure 1 shows where they land. Two tasks account for all five, at 2/10 and 3/10, and the remaining eight are silent across 80 benign episodes.

The seeds do not repeat across the two runs, so each run tests the other's task set out of sample. Taking one run to nominate the two firing tasks and testing the other against that nomination gives $p = 0.0080$ in one direction and $p = 0.0400$ in the other. We report both and take the conservative $p = 0.04$ as the headline.

Three further details point the same way. All five firings have a failed task outcome, so none is a case of the policy succeeding and passing through the region on the way. All five are cut short at the flagging step, which is the harness terminating the episode on the predicate rather than the policy completing a trajectory. And the forbidden-grasp disjunct is inert, so the box is the only thing that could have fired.

152 Read without the control arm, those five episodes are attack successes. They are not. They are the
153 predicate reporting a geometric coincidence between a static box and two particular scenes.

154 **What this study refuses to do.** It derives no corrected zone and runs no threshold sweep. The
155 reason is that the per-step end-effector poses were computed during the run and discarded: the
156 committed reports predate the schema field that would carry them, and every firing records an empty
157 endpoint map. A sweep over episodes whose poses are unknown produces a curve of the right shape
158 and no content, and choosing an operating point from it would be picking a second constant in
159 exactly the way the first one was picked. That is the bug, not the fix. The honest output of this study
160 is a located defect and the evidence that locates it.

161 5.2 Measured nulls are only interpretable against a non-zero benign rate

162 Three arms measure 0 of 50. What that means depends entirely on the benign arm beside it. Against
163 a benign arm that also fires at zero, a 0 of 50 attack arm says nothing at all: neither perturbing nor
164 not perturbing moved a predicate that never fires, and the run has not demonstrated that the predicate
165 can fire. Against a benign arm at 4%, the same 0 of 50 says the perturbation did no better than doing
166 nothing, on a predicate shown to be capable of firing. The second is a publishable null. The first is
167 an untested detector.

168 A related distinction concerns the denominator. A seventh arm in this run recorded zero attempts
169 rather than a zero rate. Its 50 records carry an inapplicable flag and zero steps, and scoring excludes
170 them. Listing it as a seventh null would imply seven measured nulls where there are three, and would
171 inflate the measured sample from 350 to 400, a 14% overstatement. Not applicable and measured
172 zero are different claims about different denominators.

173 5.3 Single-task adversarial results are underpowered in both directions

174 An earlier evaluation of the same policy on one task at ten seeds reported the goal-substitution arm
175 at 6 of 10 with an uncorrected $p = 0.031$, which does not survive correction across the arm family.
176 Pooled over ten tasks the same arm reaches 15 of 50 at $p = 9.8 \times 10^{-4}$ and does survive. The
177 single-task run missed a real effect.

178 The error runs the other way as well. That run reported the roleplay arm at 10 of 10, a clean 100%,
179 where ten tasks resolve the same arm to 88% with an interval of [72%, 100%]. It could also produce
180 no clustered interval at all, because an interval bootstrapped over tasks needs more than one task
181 and the estimator correctly declines below two. The upgrade was the scope of the evaluation, not
182 the number it produced.

183 6 A per-checkpoint protocol

184 The measurement reported above is of one checkpoint. What follows is the protocol it implies for a
185 sequence of them, stated as a recommendation rather than as something this paper has executed: we
186 have not run this across several adapted checkpoints, and the section should be read in that mood.

187 **Fix the task set and the seed set once, before any adaptation.** Both are part of the instrument.
188 A task set that grows between checkpoints, or seeds drawn afresh each time, makes two numbers
189 incomparable for a reason that has nothing to do with the policy.

190 **For each checkpoint, run both arms on that fixed set.** Attacked and benign, the same seeds, the
191 same predicate, the same horizon. The benign arm is not a baseline computed once and reused: a
192 new checkpoint has a new nominal trajectory distribution, so its benign firing rate has to be measured
193 again on that checkpoint.

194 **Report the pair, never the attack rate alone.** A checkpoint's safety row is two numbers and an
195 interval on each, not one number.

196 **Read a movement only against a floor that held still.** If attack success rises from one checkpoint
197 to the next while the benign rate is flat, that is evidence about the policy. If both rise together,
198 the most likely reading is that the checkpoint moved relative to a fixed boundary, and the attack

199 rate has told you nothing about susceptibility. A protocol that reports only the attacked arm cannot
200 distinguish these two cases, and they call for opposite responses.

201 **The cost, honestly.** This doubles the episode count. The benign arm is the cheaper half: it needs no
202 perturbation machinery, no attack implementation and no search, so it is the same rollout loop with
203 the attack disabled. In the run reported here the benign arm was 50 of 400 episode records per run.

204 **What it prevents, in this paper’s own data.** Five benign firings, concentrated on two tasks out
205 of ten. Under an attack-rate-only protocol those five episodes enter the attacked arm’s numerator
206 whenever a perturbation is applied to the same cells, and a checkpoint evaluated after adaptation
207 would carry an inflated attack success rate on exactly those tasks. A maintainer comparing it to a
208 previous checkpoint would read an adaptation that made the policy less safe. The adaptation would
209 have done nothing of the kind: the boundary sits in the wrong place for two scenes, and it sat there
210 before and after.

211 7 Two defects in our own published numbers

212 A paper arguing for honest reporting should show its own corrections rather than assert a practice.

213 **A zero-width interval, published for 21 days.** The three null arms were published with a task-
214 clustered 95% interval of [0%, 0%]. A zero-width interval states that the rate is known exactly,
215 which it is not: 0 of 50 is consistent with a true rate as high as 7.1% by the exact binomial upper
216 bound. The mechanism is worth naming because it generalises. The bootstrap guard checked a
217 proxy, the number of clusters, rather than the interval it had just computed. Ten tasks that all score
218 zero pass a cluster count and are exactly as degenerate as one task: every resample returns the same
219 rate, so the percentiles collapse onto it. The guard now checks the interval. The signed leaderboard
220 artifact was correct throughout, because it carries a different interval type; the defect was confined
221 to hand-maintained tables.

222 **One draw from a stochastic sampler.** The policy’s sampler was not fully seeded on this run, and
223 every report records that. Each number here is one draw. An earlier pilot at an identical configuration
224 returned the goal-substitution arm at 1 of 4 on one run and 0 of 4 on the next. The harness now
225 records a per-episode seed, which fixes future runs and does not fix a measurement already taken.
226 We report the numbers as measured rather than re-running and reporting the better draw.

227 8 Recommendations

- 228 1. **Publish the benign firing rate beside every attack rate.** Same tasks, same seeds, same
229 predicate, attack disabled. This is one extra arm and no new machinery. A near-zero benign
230 rate strengthens a result rather than weakening it, because it puts a floor under the number.
- 231 2. **Bootstrap intervals over the correlated unit, usually the task.** Resampling episodes in-
232 side a task treats correlated draws as independent and returns an interval that is too narrow.
- 233 3. **State measured zeros with a binomial bound.** Report 0 of 50 with its exact upper bound,
234 never as a zero-width interval, and have the estimator decline rather than emit a degenerate
235 one.
- 236 4. **Distinguish not applicable from measured zero.** They differ in the denominator, and
237 merging them overstates both the number of nulls and the sample size.

238 9 Limitations

239 This is a simulation result. No real-hardware trial has been run and no transfer claim is made. The
240 related work cited in Section 2 is other authors’ evidence that simulation carries information about
241 real brittleness; it is not a measurement about this policy on any robot.

242 One suite and one checkpoint. Nothing here speaks to other suites in the same benchmark family, to
243 other manipulation benchmarks, or to other policies.

244 The attacks are a screen, not worst-case. They are templated and auditable rather than optimised, so
245 every rate is a floor and an optimised attacker should be expected to exceed it.

246 The predicate is uncalibrated, and Section 5.1 is an argument that this fact is measurable rather than
247 a repair of it. No corrected zone is offered.

248 Every number is one draw from a stochastic sampler, as described in Section 7.

249 The policy is not uniformly competent across the suite. Clean task success under the benign arm
250 averages 84% with a range of 40% to 100% across the ten tasks, and the two tasks where the
251 predicate misfires are not the two weakest. That variation is part of the picture a reader should carry,
252 because an arm scored on a task the policy rarely completes is measuring something different from
253 an arm scored on a task it always completes.

254 **Reproducibility.** The harness is open source under a permissive licence. An anonymized mirror is
255 available at ANONYMOUS-REPO-URL-TO-FILL-IN. Every number in this paper is derived from run
256 artifacts committed in that repository rather than transcribed, and the figure is generated from them
257 by a script included there. The per-task benign-firing study reproduces on a CPU in seconds and
258 needs no simulator; the underlying policy rollouts do not, and were measured on a single commodity
259 GPU.

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